New Uzbekistan University
School of Computing

Computer Programming 1
Spring 2026
Assignment 02

Submission Deadline: Sunday 11:59 PM, May 3, 2026

Instructions

- Files must be submitted to Gradescope.com, Homework Assignment 2. Entry code: **X8JV4X.**
- Students must write **C++ code** for each of the given problems.
- All solutions should be **compiled into a single Word document**, with each problem clearly labeled and its corresponding code included.
- The assignment will be **evaluated through a viva** based on the submitted solutions. Students should be able to clearly **explain their code and logic**.
- The viva will be conducted during the **lab session in the following week**.
- There is **no restriction on using external resources**, including online materials or AI tools. However, students must **fully understand the submitted solution**, as the marks will depend on the viva performance.
- Failure to explain the submitted code during the viva may result in **reduced or zero marks**.

Note: Keep all your code files. It will be required at any time of evaluation.

Question 1. Develop C++ programs for the following

1.1 Create the following arrays in main function

1. A two dimensional array **Grades[*Number_of_students*][*Number_of_subjects*]** that stores the grades (0 ~ 100) of 6 students for 4 subjects.
2. A one dimensional array **Credit_hours[*Number_subjects*]** that stores credit hours of each subject (A course can be of either 3 or 4 credit hours)
3. A two dimensional array Grade_points i.e.
   Grade_points [*Number_of_students*][*Number_of_subjects*] that will stores the grade points of each subject for 6 students.
4. A one dimensional array **GPA[*Number_of_students*]** that stores the GPA of each student

1.2 Write the following Functions

a. A function ***Enter_Grades()*** that takes the two dimensional Grades array, number of students and number of subjects as arguments. The function ask the user to enter grades and save it in the Grades array.

b. A function ***Enter_credit_hours*** that takes the array **Credit_hours** and ask the user to enter credit hours for each subject.

c. A function ***Display_grades*** that display the values of the two dimensional array
Display function take the array and its size as arguments. (Display the grades of each student in separate line as follow)

Student	Subject1	Subject2	Subject3	Subject4
1	90	80	76	55
2	54	76	87	34
3	43	56	87	72
4	34	43	23	67
5	56	65	45	76
6	65	43	87	90

d. A function **Calculate_ Grade_points** that takes two arrays **Grades** and **Grade_points,** and **sizes i.e.,** number of students and number of subjects as arguments.

Based on the grades stored in **Grades** array, Grade points for each subject is calculate and stored in **Grade_points** array. The function **Calculate_ Grade_points** assign grades points according to Grades as follow

Grade	GPA Points
(95-100)	4.5
(90-94)	4.0
(85-89)	3.5
(80-84)	3.0
(75-79)	2.5
(70-74)	2.0
(65-69)	1.5
(0-64)	0.0

e. A function **Calculate_GPA** that takes three arrays **Credit_hours, GPA and Grade_points** and sizes i.e, number of students and number of subjects. According to grade points and credit hours this function calculate GPA for each student and store it in GPA array.

Let's say the credits hours of 4 subjects are given by CH1, CH2, CH3 and CH4, and a student got GP1, GP2, GP3 and GP4 grades in these subjects, respectively. Then the GPA for this student can be calculated is follow

GPA=(**GP1*CH1** + GP2*CH2 + **GP3*CH3** + GP4*CH4)/(CH1 + CH2 + CH3 + CH4)

1.3. File operations

a. a. Store Grades in File: Write a function: `void store_grades( int Grades[][4], int students, int subjects);`
   Create a file named grades.txt, write the grades of all students into the file, each student's grades should be written on a new line.

b. Read Grades from File: Write a function: `void read_grades(int students, int subjects);`
c. Open the file **grades.txt** , Read the grades from the file, Display them in the same format as the Display_grades function

Call these functions from main function in sequence to take data from the user, display grades and calculate GPA for each student. And then display GPA in main function.

Question 2. Implement a C++ program that perform the following operations on two 3x3 matrices. The Program should ask the user to enter the data for the two matrices and a choice of operation.

- Addition
- Subtraction
- Multiplication

For each of the above operation write a separate function. Each of these functions takes three matrices (two input matrices and a result matrix) as arguments, the operation (+ , _ , *) is performed on the input matrices and the result is stored in the result matrix.

In addition write the following function

- A function ***InputMatrices()*** Take the elements of the two matrices from the user.
  - Each matrix is 3x3
- A function ***Display()*** that display a two dimensional matrix.
  - Call this function to display the two matrices and the resultant matrix.